

2.1 Topological magneto electric effect

We have seen that topological insulators are a class of materials whose dynamic is characterised by the fact that $\theta = \pi$.

We will see that if a topological insulator is placed in an electric or magnetic field, it will exhibit many novel physical effects.

First, we place a topological insulator in a uniform magnetic field directed along the z axis. Because θ undergoes a discontinuity at the surface of the topological insulator, charge will be induced on the surface:

$$\int \sigma \cdot d\mathbf{s} = \int \nabla \cdot \vec{E} dV = \int \frac{e}{\hbar} (\nabla \theta) \cdot \vec{B} dV.$$

At the upper interface, θ jumps from π to 0 , whereas at the lower interface, θ jumps from 0 to π , so:

$$\sigma = \begin{cases} -2B & \text{(upper interface)} \\ 2B & \text{(lower interface)} \end{cases}$$

Thus, the magnetic field induces negative charge on the upper interface and positive charge on the lower interface

Alternatively, we can turn on an electric field lies tangential to the interface, we place a topological insulator in uniform electric field directed along the y ~~axis~~ axis. Similarly, because θ ~~is~~ undergoes a ~~discontinuity~~ discontinuity at the surface of the topological insulator, a surface current. $\vec{K}$ will therefore be induced:

$$\int \vec{K} \cdot d\vec{L} = \int \nabla \times \vec{B} \, ds = - \int \frac{\alpha}{2\pi} \nabla \theta \times \vec{E} \, ds$$

~~Simplify~~. Simplistically, we only consider the surface current on upper interface, so

$$\begin{aligned} \vec{K} &= -\alpha \vec{E} \cdot \hat{e}_x \\ &= -\frac{1}{2} \cdot \frac{e^2}{2\pi} \cdot \vec{E} \cdot \hat{e}_x \end{aligned}$$

The creation of a two-dimensional current which lies perpendicular to an applied electric field is called the Hall effect. The coefficient of proportionality is known as the Hall conductivity:

$$\sigma_{xy} = \frac{1}{2} \cdot \frac{e^2}{2\pi}$$

This is usually abbreviated to say that the interface of the topological insulator has Hall conductivity $\frac{1}{2}$.

The general ~~pho~~ phenomenon in which electric fields induce magnetic fields and vice versa goes by the name of the topological magneto-electric effect.

3. Weyl Semimetals.

The Weyl semimetals have nondegenerate gapless linear dispersions around band-touching points (Weyl nodes). The low-energy effective Hamiltonian around a ~~Weyl~~ Weyl node is written as:

$$H_W(\vec{k}) = Q v_F \vec{k} \cdot \vec{\sigma}$$

where $Q = \pm 1$ indicates the ~~chirality~~ chirality, v_F is the Fermi velocity and σ_i are Pauli matrices. Because the sum of ~~chiralities~~ chiralities of Weyl nodes in a system must be zero, the simplest realization of a Weyl semimetal is one with two Weyl nodes of opposite chiralities. Note that the minimal number of ~~Weyl~~ Weyl nodes of opposite chiralities in ~~Weyl~~ Weyl semimetal with broken inversion symmetry is four, while it is two in Weyl semimetals with broken time-reversal symmetry.

Now, let's consider a 4×4 continuum model Hamiltonian for two-node ~~Weyl~~ Weyl semimetals with broken time-reversal symmetry:

$$H_0(\vec{k}) = v_F (\tau_z \otimes \vec{k} \cdot \vec{\sigma} + \Delta \mathbf{I}_x \otimes \hat{\mathbf{I}} + \hat{\mathbf{I}} \otimes \vec{b} \cdot \vec{\sigma})$$

where τ_i and σ_i are Pauli matrices for Weyl-node and spin degrees of freedom respectively and Δ is the mass of 3D ^{Dirac} fermions. The term $\vec{b} \cdot \vec{\sigma}$ represents a magnetic interaction

such as exchange interaction between conduction electrons and magnetic impurities or the ~~Zeem~~ Zeeman coupling with an ~~extrat~~ external magnetic fields. Note the the Hamiltonian with $\vec{b}=0$ describes a topological or normal insulator depending on the sign of Δ (mass). Without loss of generality, we may set $\vec{b}=(0,0,b)$.

Now let's find the location of Weyl nodes. The Hamiltonian of Weyl nodes can be written as:

$$\begin{aligned} H &= v_F \cancel{\Delta} (k_x \sigma_x + k_y \sigma_y + k_z \sigma_z) + \Delta \bar{I}_x \otimes \hat{I} + b \hat{I} \otimes \sigma_z \\ &= \underbrace{v_F \bar{I}_z \otimes (k_x \sigma_x + k_y \sigma_y)}_{H_{\perp}} + \underbrace{v_F \bar{I}_z \otimes k_z \sigma_z + \Delta \bar{I}_x \otimes \hat{I} + b \hat{I} \otimes \sigma_z}_{H_{\parallel}} \end{aligned}$$

Because $H_{\perp}$ and $H_{\parallel}$ anticommute, so we can write down:

$$\begin{aligned} H^2 &= \cancel{H_{\perp}^2} + H_{\parallel}^2 + \{H_{\perp}, H_{\parallel}\} \\ &= H_{\perp}^2 + H_{\parallel}^2 \\ &= v_F^2 (k_x^2 + k_y^2) \hat{I} + \cancel{H_{\parallel}^2} \end{aligned}$$

This leads to the explicit energy spectrum:

$$E^2 = v_F^2 (k_x^2 + k_y^2) + E_{\parallel}^2$$

and $H_{\parallel}$ can be ~~split~~ split into two 2×2 blocks:

$$\begin{aligned} h_{\text{eff}} &= S(k_z \bar{I}_z + b) + \Delta \bar{I}_x \\ &= s k_z \cdot \bar{I}_z + \Delta \bar{I}_x + s b \cdot \hat{I} \end{aligned}$$

where $s = \pm 1$. So the energy spectrum of h_{eff} is given by:

$$E_{||} = \pm b \pm \sqrt{k_z^2 + \Delta^2}$$

So the Weyl node location ~~satisfies~~: ~~satisfies~~. satisfies:

$$-\sqrt{(k_x^2 + k_y^2) + b^2 \pm \sqrt{k_z^2 + \Delta^2}}^2 = \sqrt{(k_x^2 + k_y^2) + b^2 \pm \sqrt{k_z^2 + \Delta^2}}^2$$

so we can get:

$$\begin{cases} k_x^2 + k_y^2 = 0 \\ (s b \pm \sqrt{k_z^2 + \Delta^2}) = 0 \end{cases}$$

Thus, the coordinates of the Weyl nodes are:

$$(k_x, k_y, k_z) = (0, 0, \pm \frac{\sqrt{b^2 - \Delta^2}}{v_F})$$

which requires $b^2 > \Delta^2$

And then we ~~derivative~~ derive the θ term of Weyl semimetals from the microscopic four-band model. In order to describe a more generic Weyl semimetal, we add the term $\mu_5 \hat{I}_z$ to the Hamiltonian which generates a ~~chiral~~ chemical potential difference $2\mu_5$ between the two Weyl nodes. We also set $\Delta = 0$ for simplicity. So the Hamiltonian can be written as:

$$H'_0 = v_F (\hat{I}_z \otimes \vec{\sigma} \cdot \vec{k} + \hat{I} \otimes \vec{\sigma} \cdot \vec{b}) - \mu_5 \hat{I}_z \otimes \hat{I}$$

We rewrite the Hamiltonian with Gamma matrices:

$$\left\{ \begin{array}{l} \gamma^0 = \mathbb{I} \otimes \mathbb{I} \\ \gamma^i = -i\tau_i \otimes \sigma_i \\ \gamma^5 = \mathbb{I} \otimes \mathbb{I} \\ b_\mu = (\mu_5, -\vec{b}) \end{array} \right.$$

and we set $v_F \approx c = 1$, so:

$$H = -\gamma^0 \gamma^i k_i + \gamma^0 \gamma^i \gamma^5 b_i - \mu_5 \gamma^5$$

The action of the system in the presence of external electric and magnetic fields with the four potential $A_\mu = (A_0, \vec{A})$ is given by

$$S = \int dt d^3x \psi^\dagger [i\partial_t - ieA_0 - H_0(\vec{k} + e\vec{A})]$$

$$\begin{aligned} &= \int dt d^3x \bar{\psi} \gamma^0 [i\partial_t + i\gamma^0 \gamma^i \partial_i + eA_0 - e\gamma^0 \gamma^i A_i - \gamma^0 \gamma^i \gamma^5 b_i + \mu_5 \gamma^5] \psi \\ &= \int dt d^3x \bar{\psi} i [\gamma^0 \partial_0 + \cancel{\gamma^i \partial_i} \gamma^i \partial_i - ie(\gamma^0 A_0 - \gamma^i A_i) - i(\gamma^0 \mu_5 - \gamma^i b_i) \gamma^5] \psi \\ &= \int dt d^3x \bar{\psi} i (\gamma^\mu \partial_\mu - ie\gamma^\mu A_\mu - i\gamma^\mu b_\mu \gamma^5) \psi \\ &= \int dt d^3x \bar{\psi} i \gamma^\mu (\partial_\mu - ieA_\mu - ib_\mu \gamma^5) \psi \end{aligned}$$

where $e > 0$, ψ is a four-component spinor, $\bar{\psi} = \psi^\dagger \gamma^0$

Now we apply Fujikawa's method to the action. Performing an infinitesimal gauge transformation for infinite times such ~~as~~ ^{that}:

$$\begin{cases} \psi \rightarrow \psi' = e^{\frac{-i\theta(\vec{r},t)\gamma^5}{2}} \psi \\ \bar{\psi} \rightarrow \bar{\psi}' = \bar{\psi} e^{\frac{-i\theta(\vec{r},t)\gamma^5}{2}} \end{cases}$$

with $\theta(\vec{r},t) = -2\chi^\mu b_\mu = 2(\vec{b} \cdot \vec{r} - \mu_5 t)$ and $\phi \in [0, 1]$, Then the partition function Z is transformed as:

$$\begin{aligned} Z &= \int D[\psi, \bar{\psi}] e^{iS[\psi, \bar{\psi}]} \\ \Rightarrow Z' &= \int D[\psi', \bar{\psi}'] e^{iS[\psi', \bar{\psi}']} \end{aligned}$$

Because the anticommutation relation $\{\gamma^\mu, \gamma^5\} = 0$ is satisfied, so:

$$e^{\frac{-i\theta(\vec{r},t)\gamma^5}{2}} \gamma^\mu = \gamma^\mu e^{\frac{i\theta(\vec{r},t)\gamma^5}{2}}$$

The Lagrangian of $S[\psi', \bar{\psi}']$ can be written as:

$$\begin{aligned} \mathcal{L} &= \bar{\psi} e^{\frac{-i\theta(\vec{r},t)\gamma^5}{2}} \cdot i\gamma^\mu (\partial_\mu - ieA_\mu - ib_\mu \gamma^5) e^{\frac{-i\theta(\vec{r},t)\gamma^5}{2}} \psi \\ &= \bar{\psi} i\gamma^\mu (\partial_\mu - ieA_\mu) \psi + \bar{\psi} i\gamma^\mu (\partial_\mu e^{\frac{-i\theta(\vec{r},t)\gamma^5}{2}}) \psi \\ &\quad - \cancel{ib_\mu \gamma^5} \bar{\psi} i\gamma^\mu (-ib_\mu \gamma^5) \psi \\ &= \bar{\psi} i\gamma^\mu (\partial_\mu - ieA_\mu) \psi - \bar{\psi} i\gamma^\mu (1 - d\phi) ib_\mu \gamma^5 \psi \end{aligned}$$

And the Jacobian J of transformation of $D[\psi', \bar{\psi}'] = J D[\psi, \bar{\psi}]$ is given by:

$$\ln J = i \int d^4x \frac{e^2}{16\pi^2} \theta(\vec{r}, t) \star F^{\mu\nu} F_{\mu\nu}$$

We integrate with respect to the variable ϕ from 0 to 1. Due to the invariance of the partition function, we arrive at the following expression of S :

$$S = \int d^4x \bar{\psi} i \gamma^\mu (\partial_\mu - ie A_\mu) \psi + \frac{e^2}{16\pi^2} \int d^4x \theta(\vec{r}, t) \star F^{\mu\nu} F_{\mu\nu}$$

where the first term represents the trivial action of massless Dirac fermions and the second term is the θ term with ~~$\theta(\vec{r}, t)$~~
 $\theta(\vec{r}, t) = 2(\vec{b} \cdot \vec{r} - \mu_5 t)$

3.) Anomalous Hall effect and ~~chiral~~ negative magnetoresistance

let's consider the consequences of the presence of a θ term in Weyl semimetals. As we have also seen in the case of insulators with a dynamical ~~axion~~ axion field, an electric current is induced in the presence of a θ term. The induced electric current density and charge density are given by:

$$\begin{cases} \vec{j}(\vec{r}, t) = -\frac{\alpha}{\pi} (\vec{\partial} \cdot \vec{B} + \nabla \theta \times \vec{E}) \\ \rho(\vec{r}, t) = \frac{\alpha}{\pi} (\nabla \theta) \cdot \vec{B} \end{cases}$$

In the present case of $\theta(\vec{r}, t) = 2(\vec{b} \cdot \vec{r} - \mu_5 t)$, we readily obtain a static current of the form:

$$\vec{j} = \frac{2\alpha}{\pi} (\mu_5 \vec{B} - \vec{b} \times \vec{E})$$

The ~~first term~~ second term. α is the anomalous Hall effect. Let's consider a 2D plane which is perpendicular to $\vec{b}$. And we set $\vec{b} = (0, 0, b)$, which gives the 3D Hall conductivity as:

$$\sigma_{xy}^{3D} = \frac{2\alpha b}{\pi}.$$

The ~~se~~ first term is the chiral magnetic effect. The chiral magnetic effect indicates that a direct current is generated along a static magnetic field even in the absence of electric fields, when there ~~exists~~ exists a chemical potential difference $\delta\mu = 2\mu_5$ between the two Weyl nodes. However, the existence of the static chiral magnetic effect is ruled out in crystalline solid. But the chiral magnetic effect can be realized under a nonequilibrium

circumstance. When the system is driven from equilibrium by the combined effect of electric and magnetic fields. In this case of Weyl semimetal, potential difference can be generated by the chiral anomaly.

The chiral anomaly in Weyl semimetals is referred to as the electron number nonconservation in a given Weyl cone under parallel electric and magnetic fields. In which the rate of pumping of electrons is given by:

$$\frac{\partial \mu_i}{\partial t} = Q_i \frac{\alpha}{\tau} \cdot \vec{E} \cdot \vec{B}$$

where i is a valley (Weyl node) index and

$$Q_i = \int \frac{d^3k}{2\pi} \cdot \frac{\partial f_0(E_k^m)}{\partial E_k^m} \cdot \vec{v}_k^m \cdot \vec{\Omega}_k^m$$

is the chirality of the valley. Here, E_k^m is the energy of Bloch electrons with momentum $\vec{k}$ in band m in a given valley i ,

$f_0(E_k^m)$ is the Fermi ~~distrib~~ distribution function. $\vec{v}_k^m$ is the group velocity and $\vec{\Omega}_k^m$ is the Berry curvature. So the difference of the total electron number between the Weyl nodes leads to the difference of the chemical potential between Weyl nodes.

A phenomenon manifested by the chiral anomaly is a negative magnetoresistance:

$$\sigma_{zz} = \frac{\alpha}{\tau} \frac{(eB_z)^2 v_F^3}{\mu^2} \cdot \tau_{\text{inter}}$$

where μ is the equilibrium chemical potential and τ_{inter} is the intervalley scattering time. This unusual magnetoconductivity holds in the low-field limit $B_z \rightarrow 0$. So the expressing of ~~the~~ ~~first term of~~ $\vec{j}$ is understood as coming from

$$\vec{j} \propto (\vec{E} \cdot \vec{B} \tau_{\text{inter}}) \cdot \vec{B}$$

which indicates that it is a consequence of the chiral magnetic effect